
ENTERPRISE ACTIVE DIRECTORY SOC & PAM LABORATORY

End-to-End Security Monitoring, Active Directory Auditing,
Detection & Investigation

Project	Hands-on cybersecurity / SOC laboratory
Environment	Windows Server 2022 • Active Directory • Splunk • Kali Linux
Domain	corp.local
Domain Controller	DC01 • 192.168.56.101
SIEM	Splunk Enterprise • 192.168.56.1
Security Test Host	Kali Linux • 192.168.56.102
Reporting Date	29 September 2026
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Purpose

Document the laboratory as a complete SOC project: how the infrastructure was built, how identities and auditing were configured, how telemetry was collected, which controlled security scenarios were investigated, and how the evidence was turned into a Splunk monitoring workflow.

Laboratory / Portfolio Use

The completed workflow is: VirtualBox → Windows Server 2022 → corp.local Active Directory → users/groups/OUs → Advanced Audit Policy → Splunk Universal Forwarder → Splunk Enterprise → SPL detections → SOC dashboard → incident findings → response documentation.

Project objectives

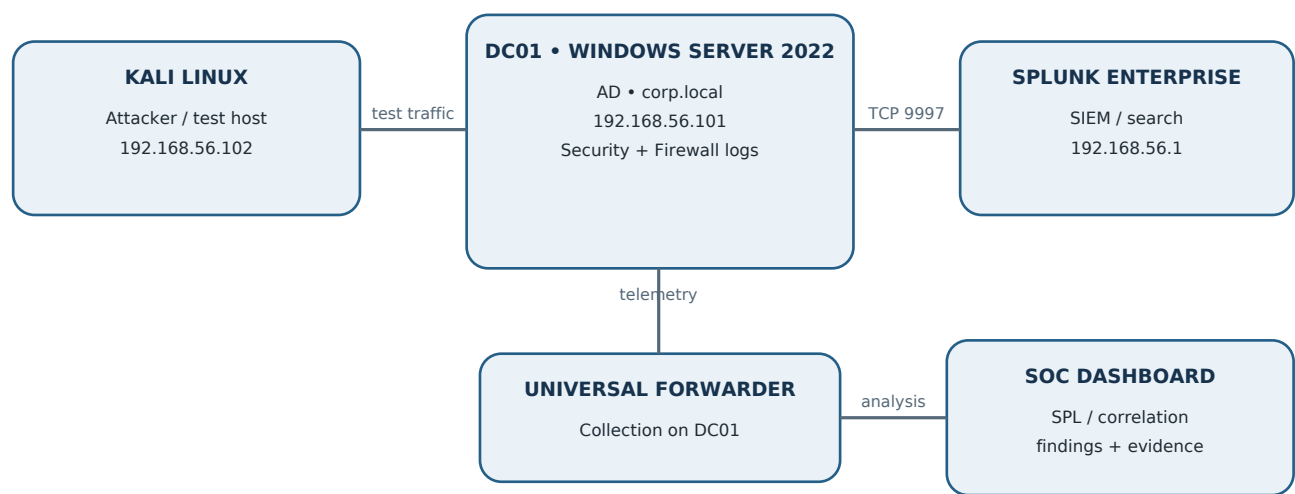
- Build and operate a Windows Server 2022 domain controller.
- Create a structured Active Directory identity model with standard, administrative, service and temporary SOC-testing identities.
- Enable Advanced Audit Policy for authentication, account, privilege, process and policy investigations.
- Forward Windows Security, System, PowerShell and Firewall telemetry into Splunk.
- Generate controlled security activity and investigate it as a SOC analyst.
- Create reusable SPL searches and an analyst-facing dashboard.
- Document findings and response actions using evidence rather than unsupported conclusions.

Project boundary

The environment is a controlled virtual laboratory. Security activity was intentionally generated for training and portfolio demonstration. Individual events are treated as investigation leads and are not automatically interpreted as proof of compromise.

2. Lab Design & Network Architecture

SOC LAB TOPOLOGY



VirtualBox host-only network • centralized telemetry • analyst workflow

Figure 1. Core laboratory topology and telemetry path.

The lab was implemented as an isolated VirtualBox environment using a host-only network. DC01 acts as the Active Directory domain controller and primary telemetry source. Kali provides controlled security-testing activity. Splunk Enterprise provides centralized search, correlation and dashboarding.

System	Role	Address
DC01	Windows Server 2022 / Active Directory	192.168.56.101
Kali Linux	Security testing / attack simulation	192.168.56.102
Splunk Enterprise	SIEM / indexing / investigation	192.168.56.1

Telemetry flow

```
DC01 Windows Security / System / PowerShell / Firewall
↓
Splunk Universal Forwarder
↓ TCP 9997
Splunk Enterprise 192.168.56.1
↓
index=main
↓
SPL searches + dashboard
```

The architecture gives the SOC analyst a single investigation point while retaining the original event context from the Windows source.

3. Windows Server 2022 / DC01 Build

The first stage was the Windows Server foundation. A Windows Server 2022 Standard Evaluation virtual machine with Desktop Experience was created in VirtualBox and used as the domain controller for the laboratory.

Build sequence

Step	Implementation
1	Create the Windows Server 2022 VM in VirtualBox.
2	Attach the host-only adapter for the isolated SOC network.
3	Assign DC01 the static address 192.168.56.101.
4	Configure the server hostname as DC01.
5	Install Active Directory Domain Services.
6	Promote the server to a new forest/domain: corp.local.
7	Verify DNS and domain functionality.
8	Create the OU and identity structure used by the SOC scenarios.

Resource profile used for the lab

Resource	Configured laboratory range
Memory	Approximately 4 GB RAM
CPU	2 virtual CPUs
Disk	Dynamically allocated, approximately 45–50 GB
Operating system	Windows Server 2022 Standard Evaluation / Desktop Experience

Verification checkpoint

Hostname: DC01
Domain: corp.local
Address: 192.168.56.101

This checkpoint established the Windows/AD foundation before telemetry collection and security testing were introduced.

4. Active Directory Domain & Identity Structure

ACTIVE DIRECTORY IDENTITY STRUCTURE

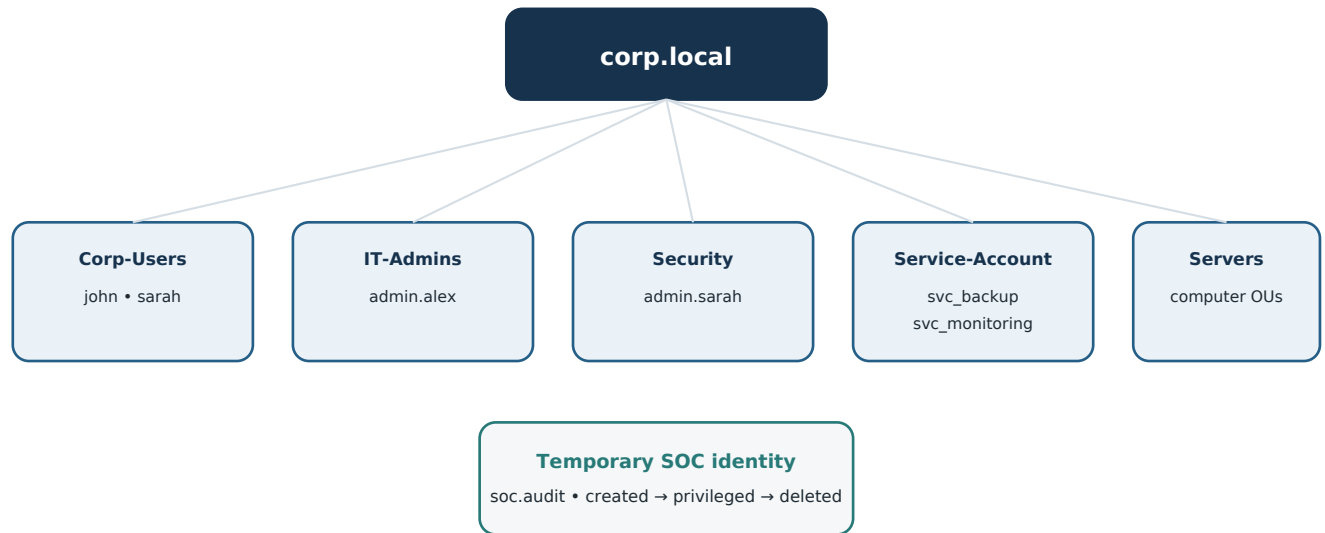


Figure 2. Active Directory OU and identity structure used in the laboratory.

OU / identity	Purpose in the lab
Corp-Users	Standard domain users: john and sarah.
IT-Admins	IT administrative identity area; admin.alex was associated with IT-Admins.
Security	Security-focused administration; admin.sarah was associated with Security-Admins and PAM-Admins.
Service-Account	Dedicated service identities: svc_backup and svc_monitoring.
Servers	Reserved structure for server computer accounts.
soc.test	SOC testing identity used during authentication testing.
soc.audit	Temporary audit identity used for account manipulation; later deleted.

Identity design

The separation of standard, administrative, service and temporary identities makes privilege-related activity easier to investigate. It also allows the SOC to distinguish ordinary authentication from administrative group changes and account lifecycle events.

5. Privileged Groups & PAM-Oriented Monitoring

The project uses Active Directory security groups to model privileged access and make changes to that access visible to the SOC. This is a PAM-oriented monitoring layer, not a claim of a full enterprise PAM product deployment.

Group	Observed role
IT-Admins	IT administrative access; admin.alex associated with the group.
Security-Admins	Security administrative access; admin.sarah associated with the group.
PAM-Admins	Privileged-access administration; admin.sarah associated with the group.
Domain Admins	Built-in high-privilege domain group; membership events were visible.
Group Policy Creator Owners	Built-in security-enabled group; membership events were visible.

Monitoring logic

Event ID	Meaning used in the project
4728	Member added to a security-enabled global group.
4729	Member removed from a security-enabled global group.

The dashboard exposes MemberName, TargetUserName, TargetDomainName, timestamp and event ID. This directly supports the SOC question: who was added to or removed from which privileged group, and when?

Production PAM systems would add approval records, ticket references, session controls and time-bound access. The laboratory demonstrates the monitoring foundation for those controls.

6. Security Auditing & GPO Configuration

Advanced Audit Policy was configured through Group Policy so DC01 generated the security events required for the investigation use cases. After configuration, gpupdate /force was used to refresh policy.

Audit area	Configuration / purpose
User Account Management	Success + Failure — account creation, changes, enable/disable, password operations and deletion.
Logon	Success + Failure — authentication activity.
Logoff	Success — session termination.
Special Logon	Success — privileged/special authentication context.
Credential Validation	Success + Failure.
Kerberos Authentication Service	Success + Failure.
Other Account Logon Events	Success + Failure.
Directory Service Access / Changes	Success + Failure.
File System / Registry	Success + Failure.
Policy Change	Audit, Authentication, Authorization and MPSSVC rule-level changes.
Sensitive Privilege Use	Enabled for privileged activity visibility.
Process Creation / Termination	Enabled for process execution monitoring.
Security State / System Extension / Integrity	Success + Failure where configured.

Design principle

The audit configuration was selected to support identity, privilege, process and policy investigations rather than collecting every possible event without purpose.

7. Splunk Enterprise & Universal Forwarder

Splunk Enterprise was installed on the Windows host at 192.168.56.1. Splunk Universal Forwarder 10.4.3 was installed on DC01 as the collection agent.

Forwarding path

DC01 Windows telemetry
↓
Splunk Universal Forwarder 10.4.3
↓ TCP 9997
Splunk Enterprise 192.168.56.1
↓
index=main
↓
SPL searches / dashboard / investigation

Configuration / check	Result
Forward destination	192.168.56.1:9997
Connectivity test	Test-NetConnection succeeded.
Forwarder status	Active forward reported to 192.168.56.1:9997.
Security input	Windows Security log enabled; index=main; renderXml=true.
Security sourcetype	XmlWinEventLog:Security

The project verified that events were actually arriving in Splunk rather than assuming that a running forwarder service meant successful ingestion. This verification step is important in a real SOC because collection failures can otherwise create false confidence.

8. Windows Defender Firewall & Controlled Security Testing

Windows Defender Firewall provided both a network control and a network-telemetry source. Allowed and blocked traffic logging was enabled, and pfirewall.log was forwarded to Splunk as windows:firewall.

```
[monitor://C:\Windows\System32\LogFiles\Firewall\pfirewall.log]
disabled = 0
index = main
sourcetype = windows:firewall
```

Controlled ICMP test

```
New-NetFirewallRule -DisplayName "SOC-Lab-Block-Kali-ICMP" `
-Direction Inbound -Protocol ICMPv4 -IcmpType 8 `
-RemoteAddress 192.168.56.102 -Action Block -Profile Any
```

Observed field	Value
Source	192.168.56.102 (Kali)
Destination	192.168.56.101 (DC01)
Protocol	ICMP
Action	DROP
Splunk sourcetype	windows:firewall
Dashboard evidence	5 displayed firewall events

Kali security-testing activities

- LDAP authentication testing against DC01.
- Controlled failed authentication used to generate 4625 evidence.
- Network connectivity testing that generated firewall telemetry.
- Command/discovery activity that produced process-creation evidence.

```
hydra -l soc.test -P hydra-test.txt ldap2://192.168.56.101
runas /user:corp\john cmd.exe
```

The current rebuilt Hydra run produced 10 attempts with no valid password discovered. The manual runas test was used to examine a directly generated 4625 event.

9. SOC Investigation — Authentication Activity

The first major investigation focused on authentication activity associated with Kali source 192.168.56.102. Event IDs 4624 and 4625 were correlated by timestamp, target username, source IP and logon type.

Time	Event	Target	Source	Type
02:27:59	4625	john	192.168.56.102	3
02:28:09	4625	john	192.168.56.102	3
02:34:23	4624	John	192.168.56.102	3
02:56:56	4625	kali	192.168.56.102	3
02:56:56	4624	ANONYMOUS LOGON	192.168.56.102	3
03:01:05	4625	kali	192.168.56.102	3
03:02:28	4625	dcgkbion	192.168.56.102	3
03:12:46	4625	john	192.168.56.102	3

Investigation finding

Multiple failed authentication events from the same Kali source were followed by successful authentication events. The sequence was therefore escalated as suspicious authentication activity for investigation. The evidence demonstrates the pattern; it does not independently establish that the domain was compromised.

Important 4625 fields observed

Field	Observed use
TargetUserName	Account affected by the authentication attempt.
TargetDomainName	corp in the investigated domain events.
Status	0xc000006d in the observed failed event.
SubStatus	0xc000006a in the observed failed event.
LogonType	2 or 3 depending on the generated event.
AuthenticationPackageName	Negotiate in the observed event.
IpAddress	Source address where present.

10. SOC Investigation — Account Lifecycle

The soc.audit scenario was created specifically to simulate account manipulation that a SOC analyst should be able to reconstruct from Active Directory telemetry.

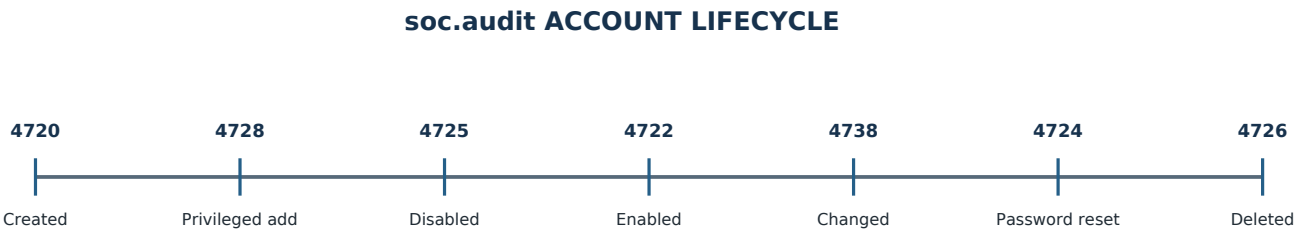


Figure 3. Account lifecycle represented from the observed event sequence.

Time	Event ID	Observed action	Account
12:38:17	4720	User Account Created	soc.audit
12:54:25	4725	User Account Disabled	soc.audit
12:54:44	4722	User Account Enabled	soc.audit
12:56:23	4738	User Account Changed	soc.audit
12:56:23	4724	Password Reset	soc.audit
12:57:10	4726	User Account Deleted	soc.audit

Privilege relationship

The account was also added to the privileged IT-Admins group and subsequently removed during the simulation. This created a combined identity + privilege narrative rather than a single isolated event.

Investigation interpretation

A rapid sequence of account creation, privilege assignment/removal, state changes, password reset and deletion is security-relevant because it can be used to establish temporary access and reduce the lifetime of an identity. In this project the sequence was intentionally created as a SOC training scenario.

11. SOC Investigation — Privileged Group Membership

The privileged group panel focuses on Event IDs 4728 and 4729. The dashboard evidence included administrative identities and built-in privileged memberships.

Member / identity	Group	Event
Admin Alex	IT-Admins	4728 — Added
Admin Sarah	Security-Admins	4728 — Added
Admin Sarah	PAM-Admins	4728 — Added
Administrator	Domain Admins	4728 — Added
Administrator	Group Policy Creator Owners	4728 — Added

Investigation fields

Field	Why it matters
MemberName	Identity whose membership changed.
TargetUserName	Account performing the group-management action, where present.
TargetDomainName	Target group/domain context.
Event time	Establishes sequence and correlation with other events.
Event ID	Distinguishes addition from removal.

PAM relevance

This is the identity-governance side of the project. In a production PAM workflow, the same monitoring would be enriched with approval, ticket, session and time-bound-access information. The lab demonstrates the underlying visibility needed before those controls can be automated.

12. SOC Investigation — Selected Process Execution

Windows Event ID 4688 was used for process creation. The raw process stream contains substantial normal operating-system activity, so the dashboard was deliberately narrowed to commands that can provide useful investigation context.

Observed process	Investigation relevance
cmd.exe	Command execution / administrative activity.
powershell.exe	Scripted administration and potential post-compromise activity.
reg.exe	Registry modification or inspection.
rundll32.exe	DLL execution / signed-binary proxy context.
wevtutil.exe	Event-log administration or clearing-related investigation lead.
taskkill.exe	Process termination.
curl.exe	Network retrieval / transfer activity.
ipconfig.exe	Network configuration discovery.

Analyst rule

These binaries are not inherently malicious. Their value comes from correlation with the executing user, parent process, command line, network activity, time and nearby security events.

The final dashboard intentionally limits this view to a maximum of ten interesting entries so the analyst sees a compact investigation surface instead of hundreds of routine Windows processes.

13. Detection Engineering & Reusable SPL

The project moved from raw event collection to repeatable searches. XML Windows events were parsed with xmlkv; rex was used where specific XML fields required explicit extraction.

Authentication failure count

```
index="main" host="DC01" sourcetype="XmlWinEventLog:Security"
| xmlkv
| search EventID=4625
| stats count
```

Authentication timeline

```
index="main" host="DC01" sourcetype="XmlWinEventLog:Security"
| xmlkv
| rex field=_raw "<Data Name='TargetUserName'>(?!<TargetUserName>[^\<]*)</Data>"
| rex field=_raw "<Data Name='IpAddress'>(?!<IpAddress>[^\<]*)</Data>"
| rex field=_raw "<Data Name='LogonType'>(?!<LogonType>[^\<]*)</Data>"
| search IpAddress="192.168.56.102" EventID IN (4624,4625)
| table _time EventID TargetUserName IpAddress LogonType
| sort _time
```

Account lifecycle / privileged groups

```
index="main" host="DC01" sourcetype="XmlWinEventLog:Security"
| xmlkv
| search EventID IN (4720,4722,4724,4725,4726,4738)

index="main" host="DC01" sourcetype="XmlWinEventLog:Security"
| xmlkv
| search EventID IN (4728,4729)
| table _time EventID MemberName TargetUserName TargetDomainName
```

Firewall detection

```
index="main" host="DC01" sourcetype="windows:firewall" "192.168.56.102"
| rex field=_raw "DROP\s+(?!<Protocol>\S+)\s+(?!<SourceIP>\S+)\s+(?!<DestinationIP>\S+)"
| table _time Protocol SourceIP DestinationIP
```

14. Splunk SOC Dashboard Design

The final dashboard was designed as the analyst interface for the lab. It combines summary counts, evidence tables and short incident-finding panels. It intentionally avoids turning every raw event into a separate visualization.

Dashboard panel	Purpose
Total Events by Data Source	Verify Security, System, PowerShell and Firewall telemetry.
Failed Logon Count	Authentication-failure context.
Successful Logon Count	Successful-authentication context.
Suspicious Authentication Activity	Kali-source authentication timeline.
Authentication Incident Summary	Concise investigation finding.
Account Management Activity	Account create/change/reset/enable/disable/delete activity.
Account Activity by Event Type	Distribution of account-event types.
Account Manipulation & Privilege Escalation	soc.audit investigation finding.
Privilege / Group Membership Changes	Privileged additions/removals.
Interesting Process Executions	Compact 4688 investigation view.
Firewall Activity	Blocked network traffic.
Network Threat Blocking Finding	Firewall investigation finding.

Dashboard snapshot — 29 September 2026

Metric	Snapshot
XmlWinEventLog:Security	58,056
XmlWinEventLog:System	3,235
PowerShell Operational	417
windows:firewall	61
Failed Logon Count	15
Successful Logon Count	7,934
Account Changed	104
Account Created	32
Password Reset	29
Account Enabled	26
Account Deleted	1
Account Disabled	1

15. Dashboard Evidence — Authentication

The first dashboard view combines telemetry availability, authentication counts and the Kali-source authentication timeline.



splunk>

Active Directory SOC Monitoring Dashboard

2026-09-29 14:33:22 India Standard Time

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Figure 4. Dashboard authentication and telemetry evidence.

Analyst use

Evidence	Purpose
Security / System / PowerShell / Firewall sources	Confirms centralized telemetry.
4625 / 4624 counts	Provides authentication context.
192.168.56.102	Allows Kali-source correlation.
TargetUserName + LogonType	Supports identity/access timeline analysis.

16. Dashboard Evidence — Account Monitoring

This view supports the account-lifecycle investigation. The incident finding provides the analyst narrative while the event table supplies the underlying Windows evidence.

Security Operations Center dashboard for monitoring and investigating Active Directory and Windows Server security events from DC01. Covers authentication activity, account management, privilege changes, process execution, and Windows D...

Incident Summary — Suspicious Authentication Activity

Suspicious Authentication Activity

Finding: Multiple failed authentication attempts (Event ID 4625) from 192.168.56.102 were followed by successful authentication events (Event ID 4624), including activity involving the john account.

Investigation: The authentication timeline was correlated in Splunk using timestamp, source IP, username, and logon type. The sequence showed unsuccessful authentication followed by successful authentication from the same source.

Assessment: The pattern was investigated as suspicious authentication activity. The events were generated as part of a controlled SOC lab simulation and do not represent confirmed compromise.

Account Management Activity

time	EventID	Event	TargetUserName	SubjectUserName
2026-09-26 12:57:10.369	4726	User Account Deleted	soc.audit	
2026-09-26 12:56:23.614	4724	Password Reset	soc.audit	
2026-09-26 12:56:23.613	4738	User Account Changed	soc.audit	
2026-09-26 12:54:44.731	4722	User Account Enabled	soc.audit	
2026-09-26 12:54:25.171	4725	User Account Disabled	soc.audit	
2026-09-26 12:38:17.069	4720	User Account Created	soc.audit	
2026-09-21 04:37:29.017	4722	User Account Enabled	soc.test	
2026-09-21 04:37:29.015	4738	User Account Changed	soc.test	
2026-09-21 04:37:29.006	4724	Password Reset	soc.test	
2026-09-21 04:37:28.940	4720	User Account Created	soc.test	
2026-09-21 04:35:42.997	4722	User Account Enabled	svc_monitoring	
2026-09-21 04:35:42.996	4738	User Account Changed	svc_monitoring	
2026-09-21 04:35:42.986	4724	Password Reset	svc_monitoring	
2026-09-21 04:35:42.923	4720	User Account Created	svc_monitoring	
2026-09-21 04:34:20.672	4722	User Account Enabled	svc_backup	
2026-09-21 04:34:20.671	4738	User Account Changed	svc_backup	
2026-09-21 04:34:20.664	4724	Password Reset	svc_backup	
2026-09-21 04:34:20.549	4720	User Account Created	svc_backup	
2026-09-21 04:31:20.873	4722	User Account Enabled	admin.sarah	
2026-09-21 04:31:20.871	4738	User Account Changed	admin.sarah	
2026-09-21 04:31:20.859	4724	Password Reset	admin.sarah	
2026-09-21 04:31:20.789	4720	User Account Created	admin.sarah	
2026-09-21 04:26:35.990	4722	User Account Enabled	admin.alex	
2026-09-21 04:26:35.990	4738	User Account Changed	admin.alex	
2026-09-21 04:26:35.986	4724	Password Reset	admin.alex	
2026-09-21 04:26:35.941	4720	User Account Created	admin.alex	
2026-09-21 04:25:05.929	4722	User Account Enabled	Sarah	
2026-09-21 04:25:05.928	4738	User Account Changed	Sarah	
2026-09-21 04:25:05.924	4724	Password Reset	Sarah	
2026-09-21 04:25:05.897	4720	User Account Created	Sarah	
2026-09-21 04:23:18.399	4722	User Account Enabled	John	
2026-09-21 04:23:18.399	4738	User Account Changed	John	
2026-09-21 04:23:18.395	4724	Password Reset	John	

splunk>

Active Directory SOC Monitoring Dashboard

2026-09-29 14:33:22 India Standard Time

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Figure 5. Account lifecycle and authentication finding evidence.

Analyst interpretation

The event table establishes the affected account and event sequence. The soc.audit activity can therefore be correlated across creation, privilege-related activity, state changes, password reset and deletion. The dashboard finding is supported by event-level evidence.

Account-event chart

The separate account-event distribution is retained in the dashboard itself; the report focuses on the underlying investigation rather than duplicating every dashboard page.

17. Dashboard Evidence — Process & Firewall

The final investigation surface combines selected 4688 process executions with Windows Firewall DROP events.

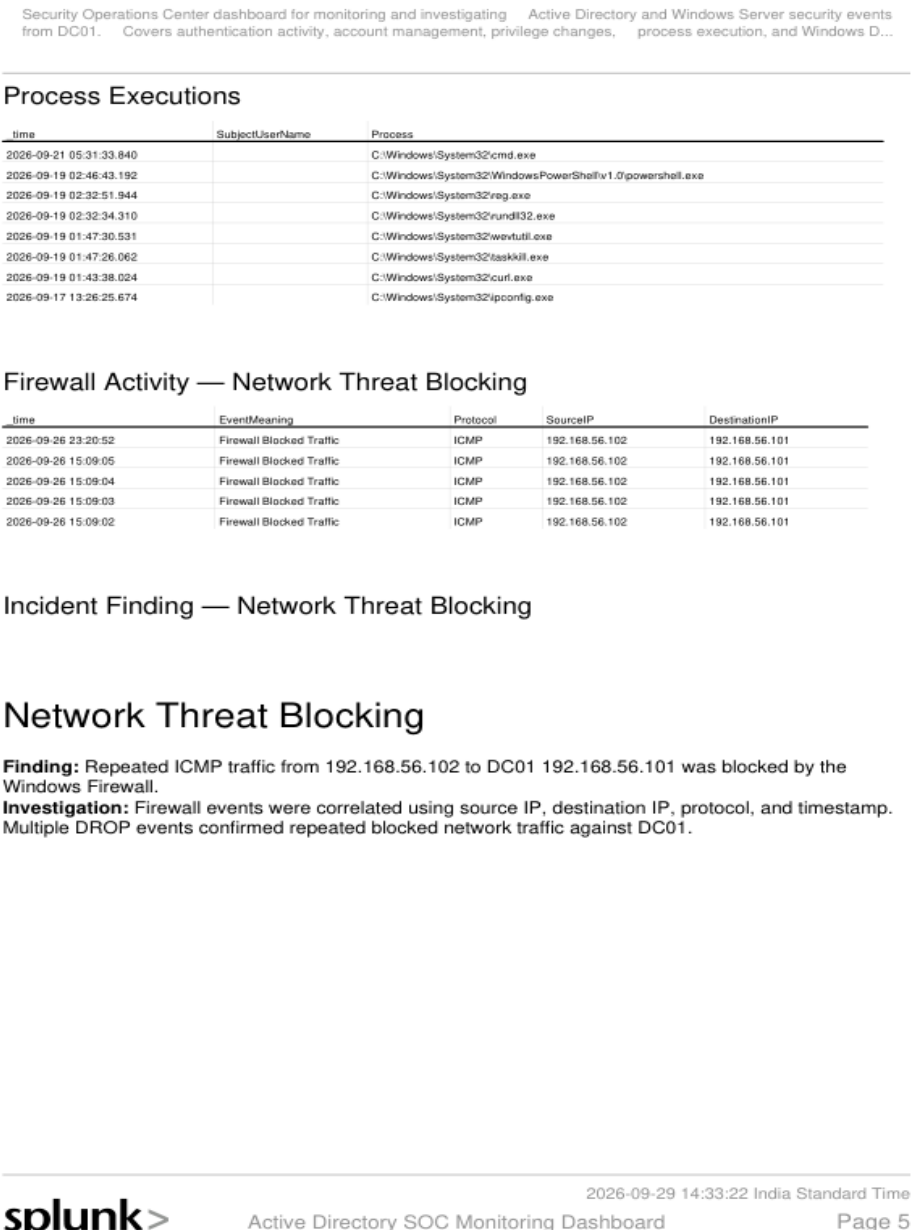


Figure 6. Selected process execution and Windows Firewall evidence.

Evidence	Observed value
Selected process view	cmd.exe, PowerShell, reg.exe, rundll32.exe, wevtutil.exe, taskkill.exe, curl.exe and ipconfig.exe.
Source	192.168.56.102
Destination	192.168.56.101
Protocol / action	ICMP / DROP
Displayed firewall events	5

Design decision

The dashboard intentionally keeps these views compact and investigation-focused rather than displaying routine endpoint activity.

18. SOC Investigation Workflow & Response



Figure 7. Practical workflow from lab construction to documented finding.

Stage	What was done
Build	Windows/AD environment, network segment, identities and privileged groups.
Audit	Advanced Audit Policy and Group Policy refresh.
Collect	Universal Forwarder for Windows Security and Firewall telemetry.
Detect	Event IDs and SPL used to isolate relevant activity.
Investigate	Source IP, username, timestamp, event, group and process correlation.
Document	Analyst findings, response approach and final report.

Response model

Finding	Response
Authentication failures	Validate source/account; correlate successful logons; consider account protection.
Privileged group change	Identify actor/member; verify authorization; remove unauthorized membership if confirmed.
Rapid account lifecycle	Review creation, privilege, password and deletion events; restrict if unauthorized.
Suspicious process / firewall activity	Correlate user, process, network and time context before containment.

19. Project Outcome, Limitations & Final Summary

Completed outcomes

Component	Status
Windows Server 2022 / DC01 + corp.local AD	Completed
OU / user / service identity structure	Completed
Privileged-group / PAM-oriented monitoring	Completed
Advanced auditing	Completed
Splunk Enterprise / Universal Forwarder pipeline	Completed
Windows Firewall logging / monitoring	Completed
Kali controlled security testing	Completed
Authentication / account / privilege investigations	Completed
Process investigation + SOC dashboard	Completed
Professional project documentation	Completed

Limitations

The environment is a controlled virtual laboratory. Process events require contextual validation; common administrative binaries are not automatically malicious. The firewall case demonstrates blocked traffic rather than successful network compromise. Dashboard metrics are point-in-time snapshots.

Next technical improvements

- Deploy Sysmon for richer process, network and file telemetry.
- Create scheduled correlation searches and Splunk alerts.
- Map detections to MITRE ATT&CK; techniques.
- Add another Windows endpoint for lateral-movement visibility.
- Introduce formal incident tickets and case management.
- Extend the PAM-oriented layer toward controlled privileged-session monitoring.

Core Event ID reference

4624	Successful logon
4625	Failed logon
4720	Account created
4722	Account enabled
4724	Password reset
4725	Account disabled
4726	Account deleted
4728 / 4729	Global-group add / remove
4738	Account changed
4688	Process created

Final project statement: The laboratory provides a reproducible entry-level SOC portfolio foundation demonstrating Windows security telemetry, Active Directory identity monitoring, Splunk investigation, controlled security testing, firewall telemetry and evidence-based incident documentation.